Week 2, Lecture 2
Beam splitter

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See Chapter 5.1 of Leonhardt, U. Essential Quantum Optics From Quantum Measurements to Black Holes (2010)
See Chapter 5.1 of Bachor, Hans-A. / Ralph, Timothy C. A guide to experiments in quantum optics (2019)

Review

What you learned last lecture…

Density operators

To describe mixed ensembles of quantum states, we use density operators:

$$\hat{\rho} = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|$$

p_i: probability of being in state $|\psi_i\rangle$

Thermal states of light

$$\hat{\rho} = \frac{1}{\bar{n}+1} \sum_n \left(\frac{\bar{n}}{\bar{n}+1}\right)^n \left|n\right\rangle \left\langle n\right|$$

Photon statistics

- **Super-poissonian**

$$\mathrm{var}\,(n) > \bar{n}$$

  - e.g. thermal state $\hat{\rho}_{\mathrm{th}}$

- **Poissonian**

$$\mathrm{var}\,(n) = \bar{n}$$

  - e.g. coherent state $\left|\alpha\right\rangle$

- **Sub-poissonian**

$$\mathrm{var}\,(n) < \bar{n}$$

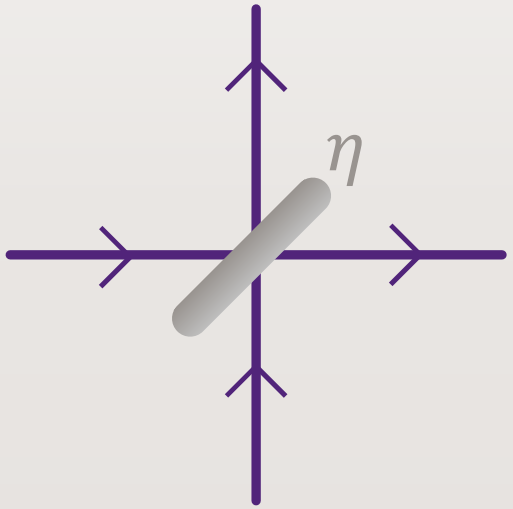
  - e.g. photon number state $\left|n\right\rangle$

Beam splitter

Introduction

- **The beam splitter is a four-port optical device**
  - Two inputs
  - Two outputs

Beam splitter diagram



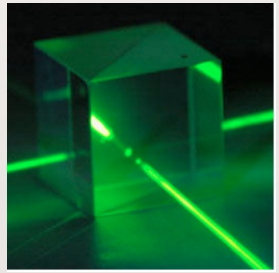
Separates beams

Different ratio η

Polarised beam splitter

Separate vertical and horizontal polarisation

Free space beam splitter



No lecture notes on the beam splitter

- See Chapter 5.1 of Leonhardt, U. Essential Quantum Optics From Quantum Measurements to Black Holes (2010)
- See Chapter 5.1 of Bachor, Hans-A. / Ralph, Timothy C. A guide to experiments in quantum optics (2019)

Beam splitter

Classical

Outgoing modes

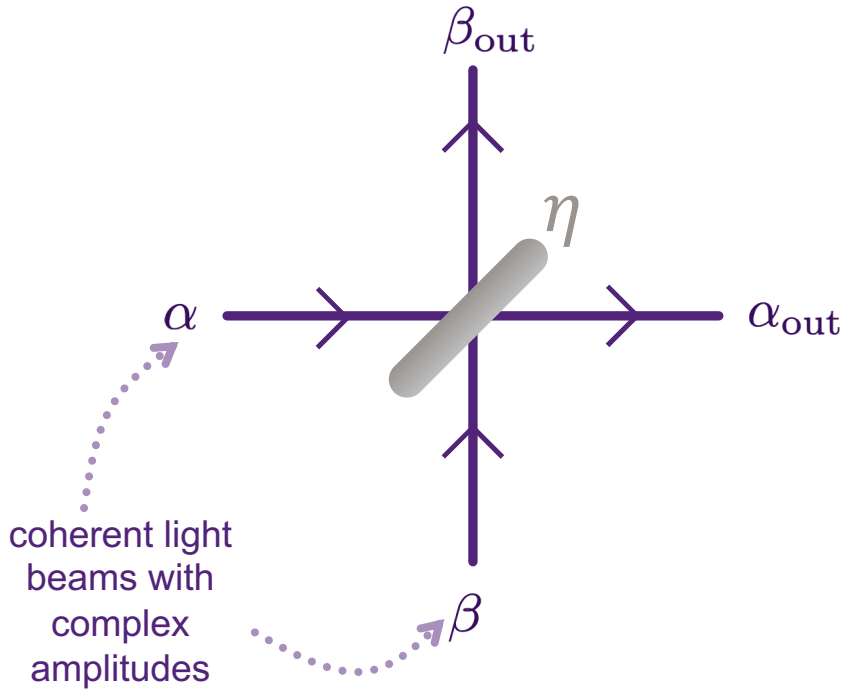
$$\alpha_{\text{out}} = \sqrt{\eta}\,\alpha - \sqrt{1-\eta}\,\beta$$

$$\beta_{\text{out}} = \sqrt{1-\eta}\,\alpha + \sqrt{\eta}\,\beta$$

The "-" sign can be placed anywhere but is essential to conserve energy since :

$$|\alpha|^2 + |\beta|^2 = |\alpha_{\text{out}}|^2 + |\beta_{\text{out}}|^2$$

lossless beam splitters conserve energy



Beam splitter

Classical

Matrix formalism

$$\alpha_{\text{out}} = \sqrt{\eta}\,\alpha - \sqrt{1-\eta}\,\beta$$
$$\beta_{\text{out}} = \sqrt{1-\eta}\,\alpha + \sqrt{\eta}\,\beta$$

$$\begin{bmatrix}\alpha_{\text{out}} \\ \beta_{\text{out}}\end{bmatrix} = B \begin{bmatrix}\alpha \\ \beta\end{bmatrix}$$

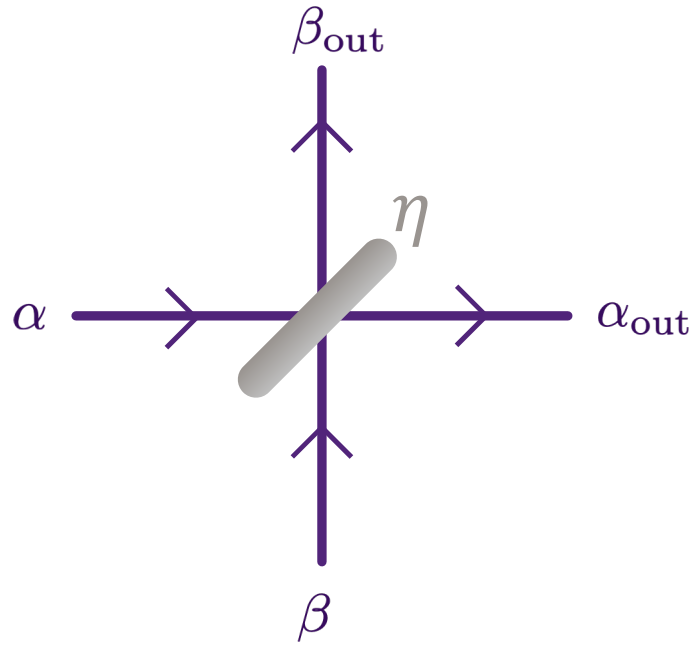
$$B = \begin{bmatrix}B_{11} & B_{12} \\ B_{21} & B_{22}\end{bmatrix}$$

$$B = \begin{bmatrix}t & -r \\ r & t\end{bmatrix}$$

t: transmissivity
r: reflectivity

Condition for energy conservation

$$t^2 + r^2 = 1$$



Here, we have used $t = \sqrt{\eta}$

$$B = \begin{bmatrix}\sqrt{\eta} & -\sqrt{1-\eta} \\ \sqrt{1-\eta} & \sqrt{\eta}\end{bmatrix}$$

Beam splitter

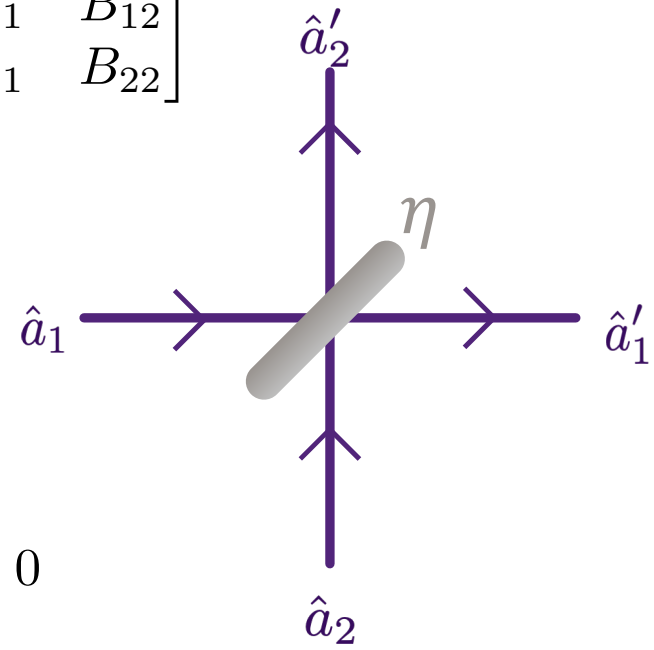
Quantum

$$\hat{a}_1' = \sqrt{\eta}\,\hat{a}_1 - \sqrt{1-\eta}\,\hat{a}_2$$
$$\hat{a}_2' = \sqrt{1-\eta}\,\hat{a}_1 + \sqrt{\eta}\,\hat{a}_2$$

$$\begin{bmatrix}\hat{a}_1' \\ \hat{a}_2'\end{bmatrix} = B\begin{bmatrix}\hat{a}_1 \\ \hat{a}_2\end{bmatrix}$$

Heisenberg picture

$$B = \begin{bmatrix}B_{11} & B_{12} \\ B_{21} & B_{22}\end{bmatrix}$$



Incoming and outgoing bosonic modes should satisfy commutation relations

$$\left[\hat{a}_\nu', \hat{a}_\mu'^\dagger\right] = \left[\hat{a}_\nu, \hat{a}_\mu^\dagger\right] = \delta_{\nu\mu}$$
$$\left[\hat{a}_\nu', \hat{a}_\mu'\right] = \left[\hat{a}_\nu, \hat{a}_\mu\right] = 0$$

$$|B_{11}| + |B_{12}|^2 = |B_{21}|^2 + |B_{22}|^2 = 1 \quad \Rightarrow \quad B_{11}B_{21}^* + B_{12}B_{22}^* = 0$$

B is unitary: $B^{-1} = B^\dagger$

Total intensity $\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2$ is invariant, ***lossless*** beam splitters conserve energy

Beam splitter

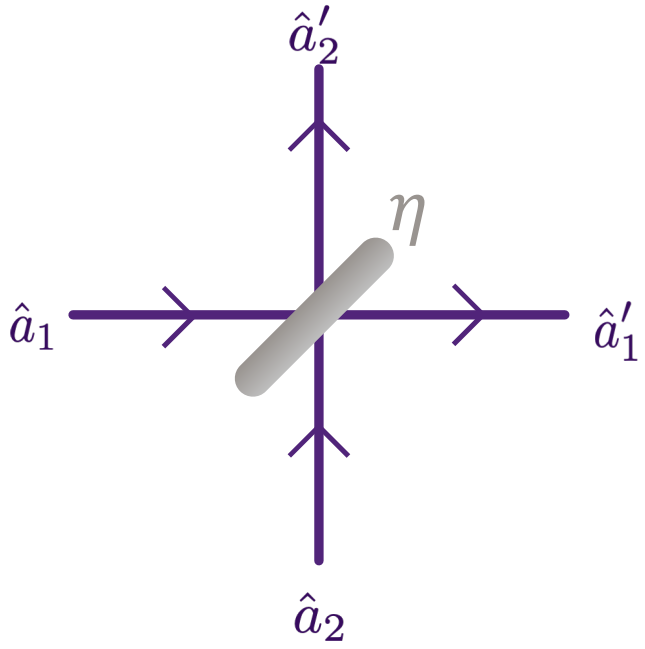
Unitary

$$\hat{U}_B = \exp\left[\theta\left(\hat{a}_1^\dagger \hat{a}_2 - \hat{a}_1 \hat{a}_2^\dagger\right)\right]$$

$$\hat{U}_B \hat{a}_1 \hat{U}_B^\dagger = \cos\theta \hat{a}_1 - \sin\theta \hat{a}_2 \qquad \hat{a}_1' = \sqrt{\eta}\,\hat{a}_1 - \sqrt{1-\eta}\,\hat{a}_2$$

$$\hat{U}_B \hat{a}_2 \hat{U}_B^\dagger = \cos\theta \hat{a}_2 + \sin\theta \hat{a}_1 \qquad \hat{a}_2' = \sqrt{1-\eta}\,\hat{a}_1 + \sqrt{\eta}\,\hat{a}_2$$

can verify this using Baker–Campbell–Hausdorf formula



$$|\psi_{\text{out}}\rangle = \hat{U}\,|\psi_{\text{in}}\rangle$$
$$\hat{\rho}_{\text{out}} = \hat{U}\,\hat{\rho}_{\text{in}}\hat{U}^\dagger$$

Beam splitter

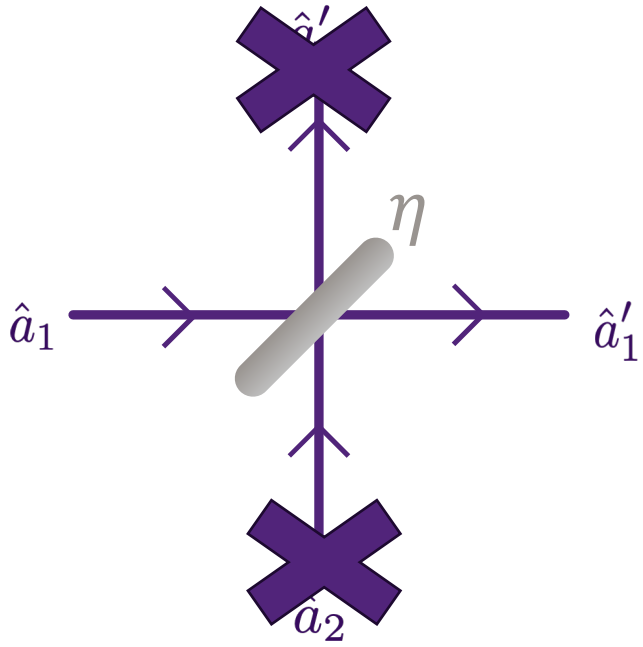
Quantum

Note that the quantum mechanical beam splitter is always a four-port device

If we ignore $\hat{a}_2$:

$$\hat{a}_1' = \sqrt{\eta}\,\hat{a}_1 - \sqrt{1-\eta}\,\hat{a}_2$$

$$\hat{a}_1' = \sqrt{\eta}\,\hat{a}_1 \quad \Rightarrow \quad \left[\hat{a}_1', \hat{a}_1'^{\dagger}\right] = \left[\sqrt{\eta}\hat{a}_1, \sqrt{\eta}\hat{a}_1^{\dagger}\right] = \eta \neq 1$$



Vacuum fluctuations from second input mode cannot be ignored

Combining Fock states

Combining n_1 photons in mode 1 and n_2 photons in mode 2:

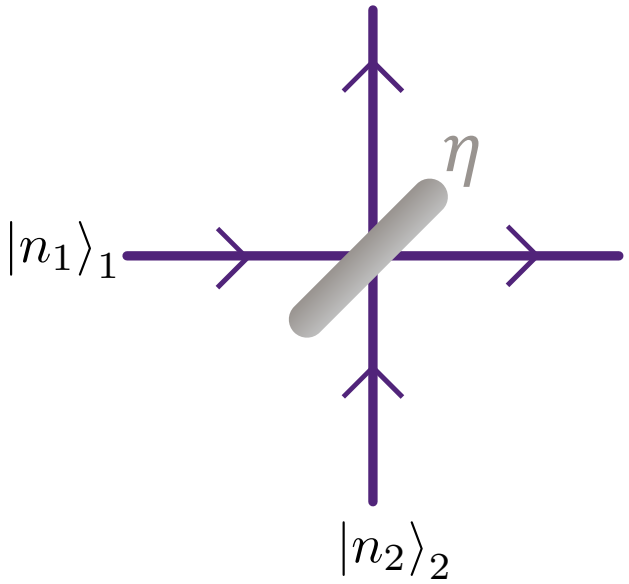
$$|n_1\rangle_1 |n_2\rangle_2 = \frac{1}{\sqrt{n_1! n_2!}} \hat{a}_1^{\dagger n_1} \hat{a}_2^{\dagger n_2} |0\rangle_1 |0\rangle_2$$

Output modes / Input modes

$$\begin{bmatrix} \hat{a}'_1 \\ \hat{a}'_2 \end{bmatrix} = B \begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \end{bmatrix}$$

Invert this transformation

$$\begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \end{bmatrix} = B^\dagger \begin{bmatrix} \hat{a}'_1 \\ \hat{a}'_2 \end{bmatrix} = \begin{bmatrix} B_{11}^* & B_{21}^* \\ B_{12}^* & B_{22}^* \end{bmatrix} \begin{bmatrix} \hat{a}'_1 \\ \hat{a}'_2 \end{bmatrix}$$



$$|n_1\rangle'_1 |n_2\rangle'_2 = \frac{1}{\sqrt{n_1! n_2!}} \left(B_{11}\hat{a}_1^\dagger + B_{21}\hat{a}_2^\dagger\right)^{n_1} \left(B_{12}\hat{a}_1^\dagger + B_{22}\hat{a}_2^\dagger\right)^{n_2} |0\rangle_1 |0\rangle_2$$

binomial coefficient

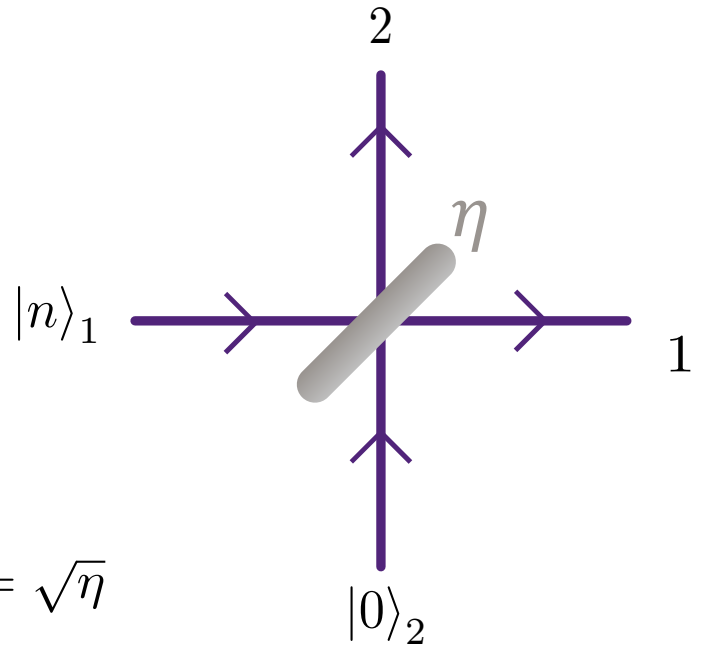
$$|n_1\rangle'_1 |n_2\rangle'_2 = \frac{1}{\sqrt{n_1! n_2!}} \sum_{k_1,k_2=0}^{n_1,n_2} \binom{n_1}{k_1} \binom{n_2}{k_2} (B_{11})^{k_1} (B_{21})^{n_1-k_1} (B_{12})^{k_2} (B_{22})^{n_2-k_2}$$
$$\times \sqrt{(k_1+k_2)!(n_1+n_2-k_1-k_2)!}\, |k_1+k_2\rangle_1 |n_1+n_2-k_1-k_2\rangle_2$$

Splitting n photons

Now, we are combining n photons in mode 1 and the vacuum in mode 2:

$$|n\rangle_1 |0\rangle_2 \rightarrow \sum_{k=0}^{n} \sqrt{\binom{n}{k}} t^k r^{n-k} |k\rangle_1 |n-k\rangle_2$$

$$B = \begin{bmatrix} t & -r \\ r & t \end{bmatrix} \qquad t^2 + r^2 = 1$$



Probability of finding k photons in mode 1 and $n-k$ photons in mode 2:

$$p_k = \binom{n}{k} \left(t^2\right)^k \left(r^2\right)^{n-k}$$

binomial coefficient

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$t = \sqrt{\eta}$$

binomial distribution

$$p_k = \binom{n}{k} (\eta)^k (1-\eta)^{n-k}$$

Single photon input

We are combining 1 photon in mode A and the vacuum in mode E:

Input

$$|1\rangle_A |0\rangle_E$$

$$|n\rangle_1 |0\rangle_2 \rightarrow \sum_{k=0}^{n} \sqrt{\binom{n}{k}} t^k r^{n-k} |k\rangle_1 |n-k\rangle_2$$

Output

$$\sqrt{\eta}\,|1\rangle_A |0\rangle_E + \sqrt{1-\eta}\,|0\rangle_A |1\rangle_E$$

↳ cannot be expressed as a product state $|\psi\rangle_A \otimes |\phi\rangle_E$

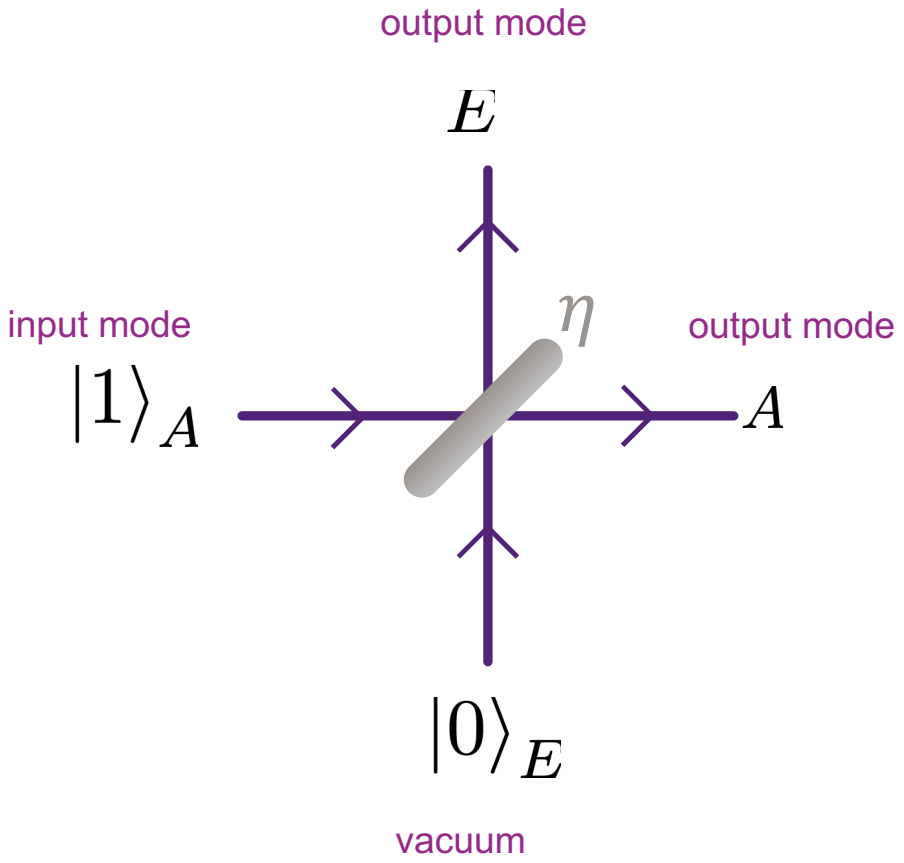
↳ Modes A and E are entangled

50:50 beam splitter

$$\eta = \frac{1}{2}$$

Bell state

$$\frac{1}{\sqrt{2}} (|1\rangle_A |0\rangle_E + |0\rangle_A |1\rangle_E)$$



Coherent states

In the Fock photon-number basis:

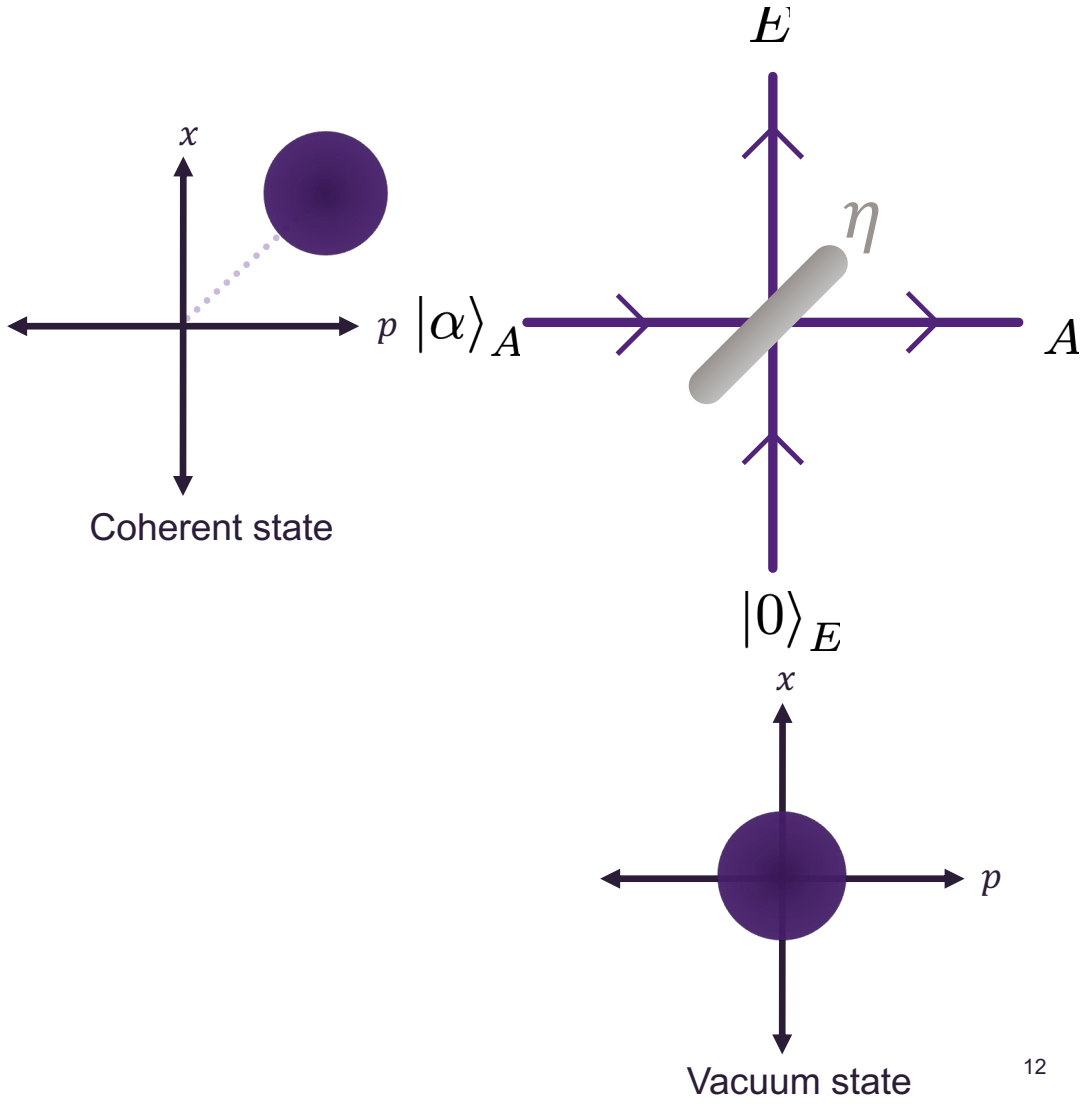
$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

Beam splitter transformation

$$|\alpha\rangle_A |0\rangle_E \rightarrow \underbrace{|\sqrt{\eta}\alpha\rangle_A |\sqrt{1-\eta}\alpha\rangle_E}$$

two coherent states in modes A and E

A single coherent state mixed with the vacuum state $|0\rangle$ does not become entangled at a beam splitter



Two coherent states

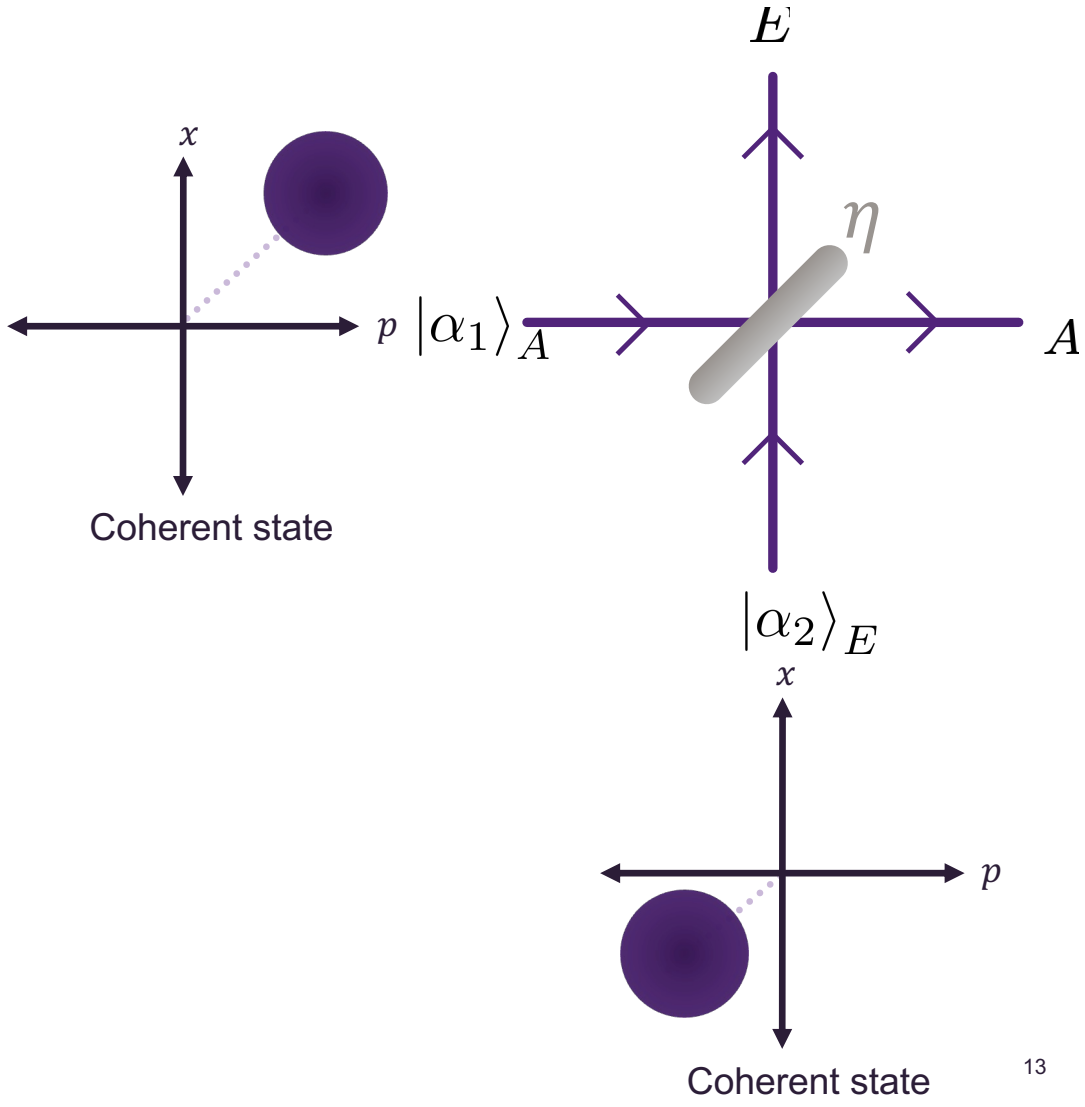
$$|\psi_{in}\rangle = |\alpha_1\rangle_A |\alpha_2\rangle_E$$

Beam splitter transformation

$$\underbrace{\left|\sqrt{\eta}\alpha_1 - \sqrt{1-\eta}\alpha_2\right\rangle_A \left|\sqrt{1-\eta}\alpha_1 + \sqrt{\eta}\alpha_2\right\rangle_E}$$

two coherent states in modes A and E

Amplitudes are transformed in the same way as classical fields



Modelling optical loss

Absorption of light can be modelled by a beam splitter

Here, we can only access and control one input and one output mode

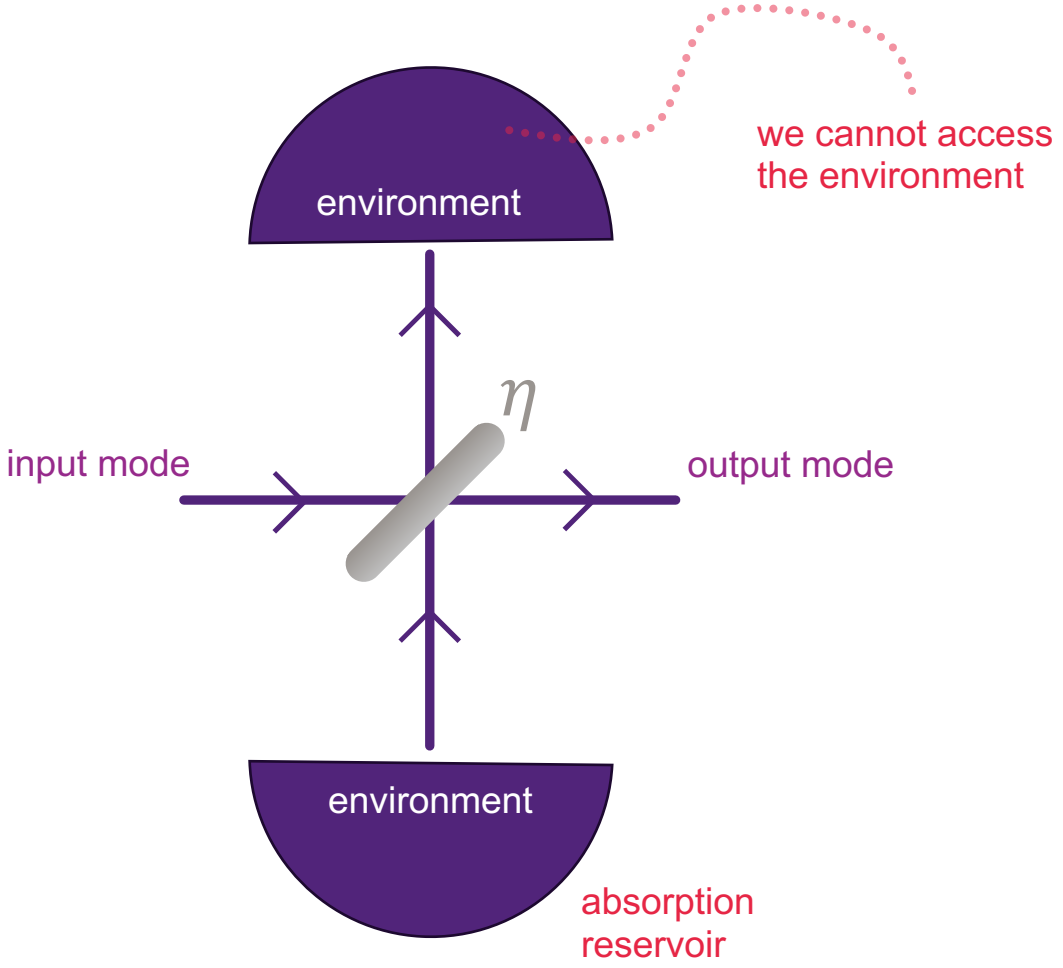
The other input mode is the absorption reservoir

The other output mode is the environment

Absorbed light is reflected into the environment

Fluctuations coming from the environment enter the outgoing light through the second input port

This can model photon absorption from optical fibre



Modelling optical loss

How do we set up the problem?

Input state on mode A

$$|\psi\rangle_A$$

Combine mode A with environment mode E using beam splitter interaction

$$|\psi\rangle_A \otimes |\phi\rangle_E \rightarrow \hat{U}_B \left[|\psi\rangle_A \otimes |\phi\rangle_E\right]$$

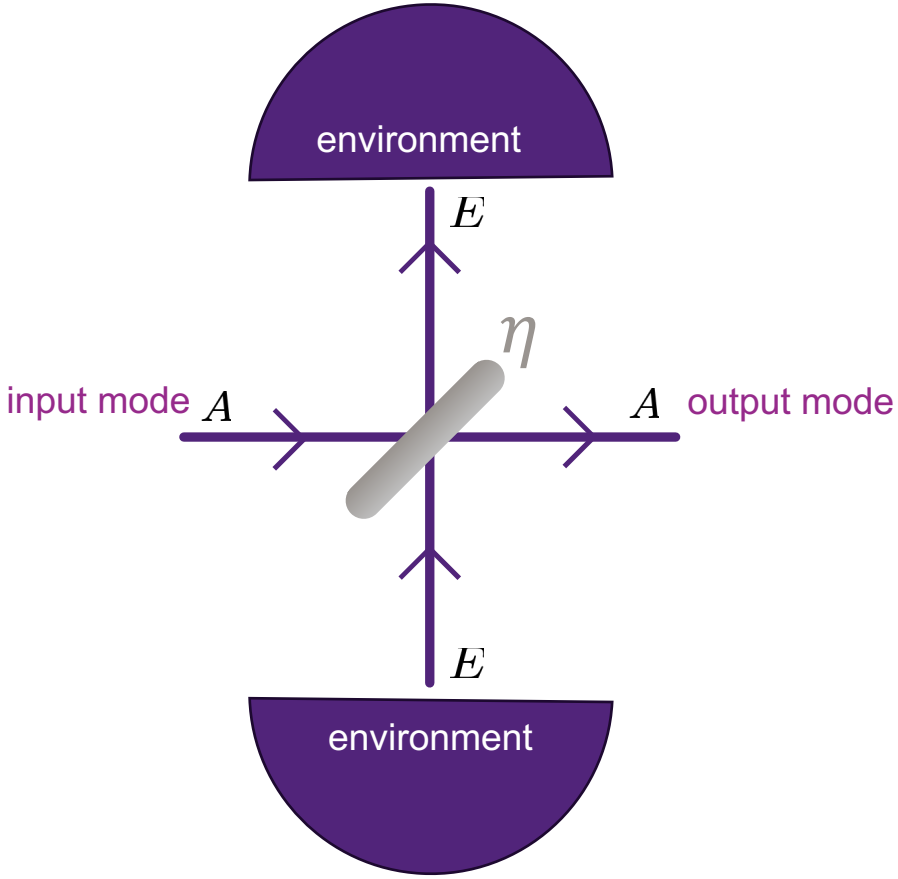
for pure loss, absorption reservoir is the vacuum

$$|\phi\rangle_E = |0\rangle_E$$

To get the reduced density matrix of our system A,
we need to trace over the evironment mode E.

Partial trace over environment mode:

$$\rho_A = \mathrm{Tr}_E \left[\rho_{AE}\right]$$



Modelling optical loss

Single photon input

$$\sqrt{\eta}\,|1\rangle_A\,|0\rangle_E + \sqrt{1-\eta}\,|0\rangle_A\,|1\rangle_E$$

- To get the reduced density matrix of our system A, we need to trace over the evironment mode E.

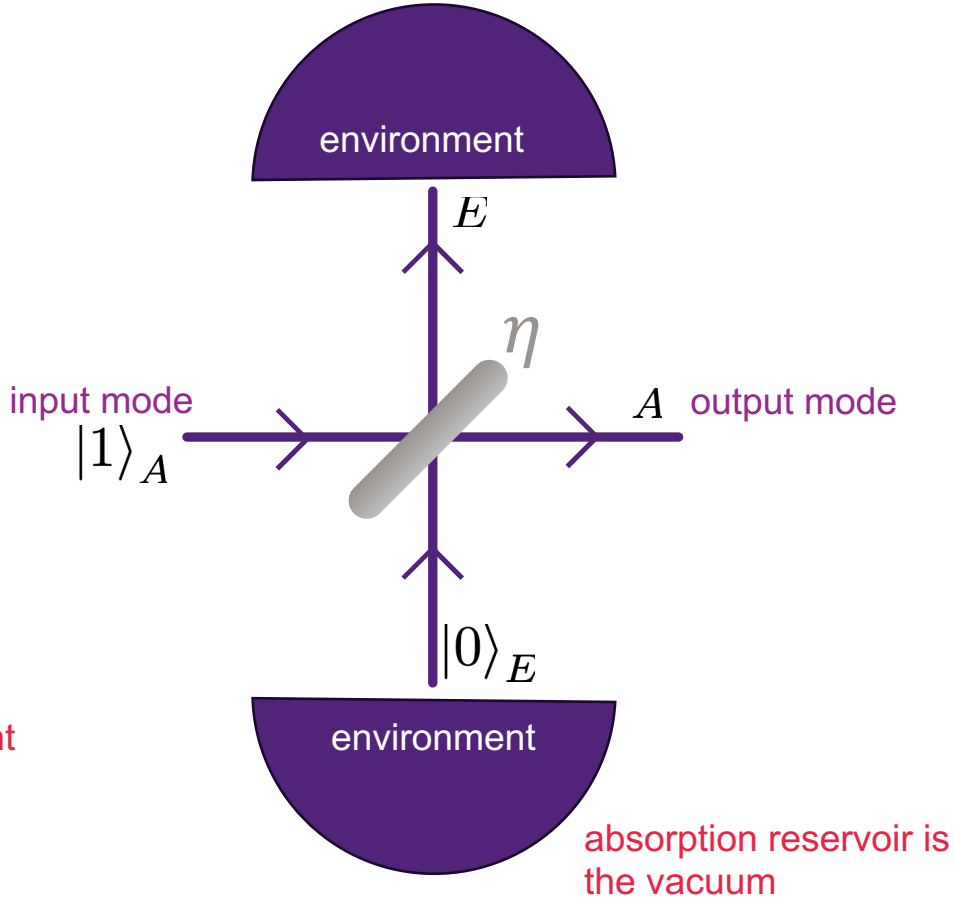
Trace over environment mode:

$$\begin{aligned}\rho_A &= \mathrm{Tr}_E\left[\rho_{AE}\right] \\ &= \eta\,|1\rangle_A\,\langle 1|_A + (1-\eta)\,|0\rangle_A\,\langle 0|_A\end{aligned}$$

single photon $|1\rangle$ transmitted probability η

single photon in environment probability $1-\eta$

- The pure state becomes a mixture of the zero photon term and the one photon term.



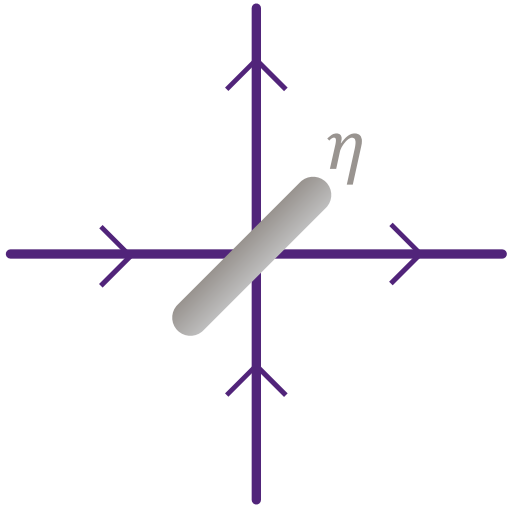
Summary

What we talked about today…

Beam splitters transform the mode operators as:

$$\hat{a}_1' = \sqrt{\eta}\,\hat{a}_1 - \sqrt{1-\eta}\,\hat{a}_2$$

$$\hat{a}_2' = \sqrt{1-\eta}\,\hat{a}_1 + \sqrt{\eta}\,\hat{a}_2$$



Splitting n photons:

$$|n\rangle_1 |0\rangle_2 \rightarrow \sum_{k=0}^{n} \sqrt{\binom{n}{k}} t^k r^{n-k} |k\rangle_1 |n-k\rangle_2$$

Beam splitters can be used to model optical loss